little extra fund, and some plastic garbage has been removed from the streets of Mombasa. When the payment is settled, the channel is closed. The worker may choose to open another channel and continue with the effort, or they may not. Their friends and neighbors may join in the effort. Thus, the blockchain-based solution is loosely coupled but agile (with quick, efficient setup and dissolution). That is the beauty of this paradigm.

7.4.3 On-chain and off-chain operations

From the discussions, comparisons, and the scenario in the preceding sections, it is evident that the interaction pattern in MPC is significantly different from that of typical web applications, in which the interaction often follows a request-response pattern. You can even say that the micropayment channel is a new paradigm that is well-suited to blockchain technology. In this new paradigm, there are off-chain and on-chain operations, like the on-chain and off-chain data you learned in chapter 6, which leads to the next design principle.

> **DESIGN PRINCIPLE 10** An important design decision in blockchain applications is to determine which data and operations are to be coded on-chain and which data and operations are to be implemented off-chain.

To better understand on-chain and off-chain operations, let’s analyze the MPC problem once more. As shown in figure 7.7, the sequence of operations provides an overview of the micropayment channel for global massive plastics cleanup. The details of the operations are as follows: